



Supporting evidences for vegetation-enhanced stormwater infiltration in bioretention systems: a comprehensive review

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Abstract

Stormwater mitigation efficiency of bioretention systems relies for a large part on their capacity to infiltrate rapidly received runoff. Within this context, the primary aim of this literature review was to clarify the vegetation influences on bioretention media hydraulic conductivity, with the ultimate goal of improving guidance on plant choice for system durability. A thorough synthesis of studies dealing with the comparison of plant species, functional types, or traits on infiltration-related processes in biofilters was achieved. Overall, results converged to a positive impact of plants on water infiltration and percolation, either under greenhouse or field conditions. In most cases, vegetation selection had a determining role in maintaining initial media infiltration rates, with in terms of improvement: turfgrass < prairie grass < shrubs < trees. Wind-induced movements of rigid foliage or stems are believed to avoid complete surface clogging. Species with thick, rhizomatous or fleshy (with maximum root diameter near the centimeter range), and tap or deep root systems could be preferred to maximize infiltration rates in permeable bioretention media. In fine-textured soils, higher specific root length, root length density, or mass density could also enhance infiltration. Root mass densities (0.1–2.2 kg.m³) were positively linked with infiltration rates in unlined systems while roots around 1 mm diameter would favor macropore-related preferential flows and increased hydraulic conductivity. Finally, implementation of high-diversity plant communities would ensure the presence of a more functionally rich vegetation community with species possessing adequate physiological adaptations (including root system architecture) to local environmental conditions for perennial cover and proper bioretention hydrological functioning.

Keywords Bioretention · Biofilter · Infiltration · Vegetation · Root traits · Hydraulic conductivity · Urban runoff · Green infrastructure

Introduction

Due to unprecedented and rapid urbanization in many parts of the world leading to dramatic increases of impervious surface areas, rainwater management has become a major issue for urban flood prevention as well as water resource preservation (Grimm et al. 2008; Roy-Poirier et al. 2010; Skorobogatov et al. 2020). In this context, the use of “nature-based solutions” (NBS) such as vegetated biofiltration systems favoring

rainwater infiltration appears as an effective way of reducing at source urban runoff while improving water quality through the retention of conveyed pollutants in soils (Roy-Poirier et al. 2010; Skorobogatov et al. 2020). Among the variety of stormwater treatment systems dedicated to runoff management, bioretention systems are green infrastructures fully integrated into the urban environment. Historically, they were developed in the early 1990s by Prince George’s County, Maryland, as “best management practice” to treat at-source stormwaters (Coffman et al. 1993; Roy-Poirier et al. 2010). These engineered systems typically consist of a layered structure comprising a mulch layer at the top, followed by a bioretention media (mix of topsoil and sand or specifically engineered substrates for pollutant retention) vegetated with ornamental plants that are often selected according to esthetic criteria or for their ability to resist alternate flooding and drying cycles (Roy-Poirier et al. 2010; Skorobogatov et al. 2020). Additional construction details may involve a diversity

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of drainage materials at the bottom, including an underdrain system to prevent water from standing in the unit for extended times in regions where native soils are of low permeability (Roy-Poirier et al. 2010). In this case, excess moisture could be directed in a storm or combined sewer system; otherwise, stormwater retention would be dominated by direct water infiltration potentially followed by exfiltration into the surrounding soil (Jennings et al. 2015; Yuan et al. 2017). As noted by Flanagan 2018, the term of “bioretention” seems to be mostly used in North America studies (Hunt and Jarrett 2004; Davis 2007, 2008; Davis et al. 2009; Lefevre et al. 2009; Traver 2011; Hunt et al. 2012, 2015) while “stormwater biofilters” (referring also to the biophysical processes allowing for water depuration in bioretention systems) are often cited in Australian studies (Dietz and Clausen 2005; Hatt et al. 2009b; Le Coustumer et al. 2012; Chandrasena et al. 2014; Flanagan 2018; Barron et al. 2019, 2020). In the present work, bioretention systems will be given a wide definition, encompassing bioretention or stormwater biofiltration systems as well as rain gardens since some authors consider these latter as bioretention cells, possessing sometimes underdrain structures (Roy-Poirier et al. 2010; Virahsawmy et al. 2014; Nocco et al. 2016; Yuan et al. 2017; Johnston et al. 2020).

As NBS, bioretention systems offer great opportunities to expand the greening of cities while providing numerous ecosystem services related to stormwater management, biodiversity support, and cultural aspects among others (Roy-Poirier et al. 2010; Skorobogatov et al. 2020). However, despite a growing body of research highlighting on bioretention systems as effective green infrastructures to manage urban runoff, field reports have regularly provided observations of some biofilter failures to cope with stormwater runoff loads (Davis et al. 2006, 2012, Hunt et al. 2006, 2012, Davis 2007, Le Coustumer et al. 2007, Hatt et al. 2009a, Li and Davis 2009, Passeport et al. 2009, Brown and Hunt 2011, Kim et al. 2012, Mangangka et al. 2015, Blecken et al. 2017, Jiang et al. 2017, Beryani et al. 2021). In a former survey assessing the media hydraulic conductivity of Australian biofilters (through the measurement of saturated hydraulic conductivity— K_{sat}), Le Coustumer et al. (2007) showed that 43% of devices were below the minimal infiltration rate of 50 mm/h recommended by Australian biofilter guidelines. According to the authors, this would have been mainly due to clogging phenomena following fine particulate matter or sediment deposition onto the surface or into the pore spaces of bioinfiltration media, as well as inadequate design choices regarding the specifications of the original filter media materials (Le Coustumer et al. 2007). Clogging may also occur in bioretention systems equipped with an underdrain structure, where orifice outlet controls must allow sufficient long residence times for contaminant removal (Sileshi et al. 2010). Indeed, these later are usually small (< 10 mm) and consequently prone to clogging

(Sileshi et al. 2010). Overall, from ~30 to 40% of stormwater infiltration facilities including biofilters may fail in their primary objective of stormwater infiltration within a few years only following their implementation (Lindsey et al. 1992; Le Coustumer et al. 2007, 2009). Despite the fact that several kinds of bioretention media have been engineered during the last decades, involving coarser media to reduce clogging as well as selected filter materials for pollutant retention (Pitt et al. 2010; Skorobogatov et al. 2020; Kuoppamäki et al. 2021), design errors and/or insufficient pre-treatment are additional factors potentially leading to common failure in bioretention cells (Beryani et al. 2021). Recently, a biofilter survey conducted in Sweden indicated that ~40% of the studied facilities were not operating optimally due to low hydraulic conductivity rates with noticeable sediment accumulation within biofilters (Beryani et al. 2021).

Besides esthetic issues, disruption of stormwater infiltration could result in higher detention times and thus, to more frequent overflows as well as reduced contaminant removal, ultimately leading to discharges of polluted water that would no longer meet regulatory thresholds (Lindsey et al. 1992; Le Coustumer et al. 2007). When biofiltration systems are oversized, their stormwater detention storage capacity may sometimes compensate for decreased media hydraulic conductivity (Lecoustumer et al. 2009). However, this is rarely the case in high-density urban settings where small bioretention cells possessing underdrains are often predominant. Moreover, even when infiltration rates are maintained over years (Spraakman et al. 2020), bypass flow as a function of infiltration rates may alter stormwater treatment performances, especially for dissolved contaminants for which targeted bioretention design is still required (Hunt et al. 2012; Lange et al. 2020; Sprakman et al. 2020; Rodgers et al. 2022). Besides, stormwater runoff management failure could be worsened in the upcoming years considering climate change leading to more frequent heavy rainfalls as well as potentially prolonged dry periods (Bastin et al. 2019). This will result in strain increases on bioretention systems including vegetation communities.

Numerous monitoring studies in the field of ecological and agricultural sciences have documented relationships between the presence of plants and substrate hydraulic properties in vegetated soils (Archer et al. 2002; Lu et al. 2020 and references therein). Lu et al. (2020) recently reviewed the literature dealing with root-induced modifications of soil hydraulic properties, following soil column and soil core studies that were undertaken either in laboratory or under outdoor (natural) conditions. Soil hydraulic properties were assessed from surface to a maximum column depth of 75 cm (with only one study reaching up to 115 cm depth) and for a variety of vegetation cover (grass, shrubs, and trees) (Lu et al. 2020). Overall, they showed that roots may substantially modify the saturated hydraulic conductivity (K_s) of soils, ranging from –143 to 1085%.

More precisely, it appeared that at low density, fine roots (diameter < 1 mm) would block soil pores, therefore reducing saturated water content (θ_s) and K_s (Lu et al. 2020). At high density, fine roots would drastically decrease macropores (> 30 μm) while increasing pore volume and thus θ_s but reducing K_s (Lu et al. 2020). By contrast, coarse root development (diameter > 2 mm) would lead to small-scale soil compaction, resulting in soil particle re-orientation with macropore development, and ultimately to higher θ_s and K_s (Lu et al. 2020). Root decay could greatly enhance K_s (up to two orders-of-magnitude) by favoring continuous macropore formation. Finally, positive root influences on K_s could be anticipated for fine-textured soil. The opposite trend (both for K_s and θ_s) could be expected for coarse-textured soils while their water retention capacity would be increased. Such narrow and complex relationships between root development and soil hydraulic properties usually vary depending on plant functional type and their biological characteristics which are directly involved into ecosystem functioning (i.e., functional traits), especially their root system architecture and morphology when focusing on water infiltration (Archer et al. 2002; Lu et al. 2020; Mair et al. 2022). Well-known conceptual models were described, for example, in Archer et al. (2002), to explain the higher infiltration rates observed for instance under a shrub vegetation cover when compared to a grass cover in natural ecosystems in arid southern Spain regions. Accordingly, larger root diameters (and volumes) at greater soil depth with perennial shrubs compared to perennial grasses (characterized by finer fascicular roots at shallow depths) would be responsible for larger belowground root channels and macropore formation (i.e., pores greater than 300 μm for the creation of preferential flows (Jarvis 2007)) in these natural arid ecosystems (Archer et al. 2002). Consequently, water infiltration, percolation, storage, and availability into deeper soil layers, also potentially associated with increased dissolved nutrients in the soil solution, would be enhanced (Archer et al. 2002). Finally, the observed variability in root depths in natural mixed-species ecosystems may help reducing interspecific competition while promoting a better water distribution and use including nutrient resource availability at the plant community level (Archer et al. 2002). To what extent such naturally occurring processes based on “niche complementarity” principle and leading to enhanced biodiversity-ecosystem functioning could occur in bioretention systems would deserve further investigations, as suggested by Levin and Mehring (2015). Besides, successful transposition of ecological principles and theories to bioretention design and functioning would also depend on the implementation context and on the construction characteristics of bioretention systems. Indeed, enhanced infiltration and drainage could not be always desirable if bypassing the filtration function of bioretention media, while in some cases (e.g., high flow

media and small size systems, below 5% of contributing area), infiltration could not be the primary flow path at all. Moreover, media depth and composition may strongly influence plant development and subsequent multicomponent interactions (Skorobogatov et al. 2020).

With this in mind, a thorough and comprehensive assessment of the scientific investigations conducted so far on the influence of plant species on stormwater infiltration (and percolation) appears as a prerequisite to clarify the current state of knowledge and to help in designing more sustainable bioretention facilities that would meet operational demands for improved urban runoff management. Extended to the context of urban green infrastructures, the ecological principles underlying the functioning of common vegetated terrestrial ecosystems would probably help to improve the design of NBS like bioretention systems, especially the selection and management of the introduced plant species in order to optimize their hydrological functioning and ecosystem services (Levin and Mehring 2015). Therefore, this review summarizes the specific illustration and understanding of vegetation effects on bioinfiltration-related processes. First, general observations regarding the role of abiotic factors (i.e., others than plant selection) on stormwater infiltration performances are briefly described. Evidences are then reported on the role of plants to enhance media hydraulic conductivity properties depending on the plant functional type. A special emphasis was made on root functional traits (and presumably associated rhizosphere processes) that have been shown to increase hydraulic conductivity of biofilter media, confirmed by field measurement when data were available (mainly infiltration tests, generally without any specific distinction between macropore and matrix flow contribution). The influence of spontaneous plant species on the maintenance of media hydraulic conductivity are also discussed. Relying on this literature synthesis, concluding remarks on bioretention design and vegetation selection as well as future research prospects are proposed to improve the long-term effectiveness of bioinfiltration systems.

Search methodology and analysis methods

This literature review was written as a narrative review dedicated to researchers and practitioners, offering a critical assessment of the works that have been conducted so far to clarify the influence of vegetation on water infiltration in bioretention systems. However, in order to identify and synthesize the state-of-the art knowledge on this topic, we initially followed a systematic assessment of the literature using well-known bibliographic database search and retrieval systems. Thus, a preliminary literature search (focusing on peer-reviewed articles) was first performed in the multidisciplinary database Sciverse Scopus (restricted

to English-language papers), using the keywords “rain garden,” “raingarden,” “bioretention,” “bioinfiltration,” or “biofiltration device” (singular and plural). This search resulted in 1553 references including 835 peer-reviewed articles. A review of those papers (based on title and keywords, abstract, or full manuscript analysis when needed) enabled to retain only 11 documents aiming at investigating the influence of plants (including functional traits) on the media hydraulic conductivity of “bioretention systems,” focusing on “bioretention,” “biofilter,” or “rain garden” as the studied systems according to our search keywords. Obviously, there is a considerable literature on how plants may increase field infiltration rates, see for instance Lu et al. (2020) and references therein, as well as (Muñoz-Carpena et al. 1993; Gilker et al. 2002; Blanco-Canqui et al. 2004a, b; Rachman et al. 2004; Seobi et al. 2005; Sharma et al. 2006; Udawatta et al. 2006) for vegetative buffer strips allowing for runoff, soil and sediment loss reduction along with water infiltration improvements in natural or agroforestry systems. By contrast, the present work is aimed at focusing more particularly on the plant influences in bioretention systems, which are nature-based solutions dedicated to urban stormwater management, usually possessing specifically engineered media and receiving varying amounts of stormwater runoff. After an in-depth analysis of all the recovered articles from the literature search, a restricted number of conference papers, scientific investigation report, or PhD thesis dissertation (which were not retrieved by the bibliographic database search but still related to the topic of vegetation influence on water infiltration in bioretention systems) were added, leading to 16 research documents (Table 1). Although we acknowledge that the present literature review is not exhaustive, we believe the generic set of search terms as well as the “systematic” procedure of article analysis would have allowed capturing the most influential papers on this topic. Moreover, as previously mentioned, a number of additional papers dealing with the effects of grass barriers and vegetative filter strips on runoff reduction have been also considered that improved the overall findings on plant functional type and traits upon runoff management and infiltration in vegetated areas. Despite a growing interest in the last decades for NBS and their provided ecosystem services, it appeared that little work to date has been conducted compared to agronomical or ecological domain to thoroughly characterize the effects of plants (and more specifically their associated root functional traits) on the infiltration performances and the evolution of hydraulic conductivity of bioretention systems. However, as it will be illustrated in the following sections, a number of common threads have emerged from experiments dealing with bioinfiltration performance evaluation in bioretention systems, involving either abiotic or biotic (vegetation-related) factors, also paving the way for future research investigations.

Results and discussion

Major abiotic factors affecting stormwater infiltration in bioretention facilities

Stormwater load-related bioretention media compaction

Monitoring of three Australian biofilters by Hatt et al. (2009a) confirmed their efficiency to attenuate stormwater runoff peak flow, with an average reduction level of 80% (i.e., from 37 to 96% peak flow lowering depending on rainfall characteristics—intensity and duration). These results regarding the effects of vegetative systems on runoff reduction were congruent with those previously reported for grass and hedge barriers and vegetative filter strips in agricultural lands (discussed in more details in the section “Evaluation of the influence of plant functional type and traits quantitative stormwater management; a. General field study considerations regarding runoff treatment with vegetative buffer strips”). Meanwhile, this field study clearly illustrated biofilter capacity to capture and retain runoff, with up to 33% of the incoming runoff volume retained in a lined bioretention system prior to evapotranspiration losses (Hatt et al. 2009a). Nevertheless, progressive and marked decreases of the hydraulic conductivity were observed in these devices as early as a few weeks following their implementation (Hatt et al. 2009a). In one of the “undersized” green infrastructure (i.e., bioretention surface area representing only 1% of the impervious catchment area), the initial infiltration rate of 300 mm/h dropped to 180 mm/h (targeted infiltration performance) just after 2 weeks, to finally reach 20 mm/h after 6 months (Hatt et al. 2009a). Such alterations of biofiltration media hydraulic conductivity were consistent with those previously reported in other bioretention ponds (Lindsey et al. 1992; Le Coustumer et al. 2007), as well as biofilter column experiments (Le Coustumer et al. 2007, 2009; Bartens et al. 2008). Indeed, Le Coustumer et al. (2007) reported an average decrease of infiltration rate of 66% (i.e., from 521 to 176 mm/h) following 14 weeks of greenhouse biofiltration column testing. Further monitoring eventually lead to a water-infiltration reduction factor of ~3.6 over a 72-week period, with a final infiltration rate of 51 mm/h on average (either in planted or unplanted units) compared to 186 mm/h at the beginning of this study (Le Coustumer et al. 2012). In fact, it appears that natural bioretention media compaction is likely to occur in the weeks to months following device implementation, in relation with successive stormwater inflow loadings (Le Coustumer et al. 2007, 2012; Hatt et al. 2009a). Media compaction may be also aggravated when pedestrian passage zones exist in the largest infrastructures (Virahsawmy et al. 2014). All together, these observations raise the issue of whether design choices,

Table 1 List of research documents (regular articles, research report, and PhD dissertation) dealing specifically with the assessment of the influence of plant species, functional type, or traits on bioretention media hydraulic conductivity

References	Experimental design	Plant species	Main findings
(Barron et al. 2019, 2020)	-Bioretention column experiment: vegetated and non-vegetated PVC columns designed according to biofiltration guidelines for Melbourne, Australia (240 mm diameter, ~1000 mm total height, 4 layers of washed sand/washed sand + cedar mulch/coarse sand/gravel) -Dual-mode stormwater-greywater biofilter devices	- 3 ornamental understory species (<i>Strelitzia nicalai</i> , <i>Phormium tenax</i> , and <i>Canna x generalis</i>), 3 climbing species (<i>Pandorea jasminoides</i> , <i>Parthenocissus tricuspidata</i> , and <i>Lonicera japonica</i>) -1 traditional species of Australian biofilter: <i>Carex appressa</i> -1 common wetland species: <i>Phragmites australis</i>	-Confirmed benefits of dual-mode biofilters to maintain healthy vegetation as well as biofiltration system performances with effective removal of total suspended solids (83–94%), biochemical oxygen demand (86–93%), and some heavy metals (lead 96–98%, copper > 93%); critical plant species selection for nitrogen and phosphorus removal (highest performances with <i>Carex appressa</i> and <i>Canna x generalis</i>) -Enhanced stormwater infiltration rate due to rhizomes of <i>Canna x generalis</i> and fleshy roots of <i>Phormium tenax</i> , but constrained large root volumes (<i>Carex appressa</i> , <i>Canna x generalis</i>) impeded infiltration after 3-year experiments -In average 63% more water infiltration (up to 153%) with black oak or red maple versus unplanted control over one growth season -27 times more infiltration with red ash (following 2 years of plantation) -Runoff reduction rate of 70–83%, 100%, and 68–78% respectively with a bioswale, infiltrating trench, or grass strip -Increased infiltration with decreasing clay/silt soil content -Increased infiltration performances under prairie grass compared to turfgrass, with respectively 37% and 10% runoff reduction rate in a silty clay soil -Infiltration test (single ring infiltrometer) indicating higher saturated hydraulic conductivity (K_s) in vegetated areas than in bare soil -Increased infiltration during the vegetative growth period - <i>Phalaris arundinacea</i> was the best performing species to limit clogging due to its rhizomatous propagation mode with horizontal and underground stems High root mass density, root surface density, root diameter, and root volume density were identified as root functional responsible for enhanced infiltration
(Bartens et al. 2008)	I-BMP (innovant best management practice for stormwater) tree pit	-Black oak (<i>Quercus velutina</i> Lam.) -Red maple (<i>Acer rubrum</i> L.) -Red ash (<i>Fraxinus pennsylvanica</i> Marsh)	
(Bloorchian et al. 2016)	-Modeling of 3 kinds of vegetated biofilters: bioswale, infiltrating trench, and grass strip -Infiltration modeling using “Green-Ampt” from USEPA-PCSWMM (Personal Computer Storm Water Management Model, GIS version of SWMM)	-Turfgrass -Prairie grass	
(Gonzalez-Merchan et al. 2014)	-Vegetated infiltration basin situated at the outlet of a detention/retention basin -Total area of the basin: 8000 m ² , with a volume of 61,000 m ³ ; it has been functioning for more than 30 years but scrapped in 2004 (sediments and the topsoil totally removed)	Spontaneous vegetation (typical invasive wetland species): - <i>Phalaris arundinacea</i> - <i>Polygonum mite</i> - <i>Rumex crispus</i>	
(Hart 2017)*	15 bioretention facilities of similar design and age (about 3.5 years old, 19 m ² , ratios 6.6%)	-10 facilities mainly covered by rushes (<i>Juncus patens</i>) and some young trees -5 facilities with a rush dominant assemblage and 5 others with a <i>Carex</i> dominant assemblage	

Table 1 (continued)

References	Experimental design	Plant species	Main findings
(Hatt et al. 2009a, b)	3 biofiltration basins (Australia) of respectively 1%, 2%, and 5% of their catchment area	Autochthonous vegetation: - <i>Carex appressa</i> , <i>Carex tereticaulis</i> - <i>Lomandra longifolia</i> - <i>Isolepis nodosa</i> - <i>Coleocephalus lacteus</i> - <i>Juncus</i> spp.	-Overall efficiency of all facilities to reduce peak flow rates (80% reduction) -Undersized biofilter (1% of the catchment area) leading to 33% on average of stormwater retention volume but 11 rainfall episodes (for a total of 28) led to overflows -Development of a moss layer (due to the installation of a “hothouse” in winter to accelerate plant growth) in addition with compacted media (because of hydraulic loads) decreased infiltration while moss removal allowed for enhanced infiltration coinciding with vigorous vegetation growth -Greatest reduction in drainage volume observed in presence of shrubs and prairie grass (in relation with evapotranspiration and media storage volume replenishment) -By contrast, accelerated drainage reported with turfgrass -Plant-induced changes in antecedent soil moisture to be considered in the hydrological functioning of rain gardens
(Johnston et al. 2020)	12 rain gardens (outdoor mesocosms) arranged according to a complete randomized block design to test the influence of 4 vegetative treatments on hydrology and soil development	-Control treatment: no vegetation -Turfgrass treatment: <i>Poa pratensis</i> -Prairie treatment: mix of 19 species (Asclepiadaceae, Cyperaceae, Fabaceae, Lamiaceae, Poaceae, and Scrophulariaceae) -Shrub treatment: mix of 6 species (Aquifoliaceae, Caprifoliaceae, Cornaceae, Rosaceae, and Salicaceae)	-43% of bioretention facilities were below their targeted infiltration performances according to Australian guidelines, i.e., media hydraulic conductivity ranging from 50 to 200 mm/h -66% of decreased hydraulic conductivity reported on average in bioretention columns after 14 weeks of experiment -Higher infiltration (but nonsignificant due to a high variability) in planted columns showing a dense vegetation growth (e.g., <i>Carex appressa</i>) compared to sparse plant cover (<i>Melaleuca ericifolia</i>) or shallow root system (<i>Microlaena stipoides</i>) -Negative impact of a ratio of [biofilter surface/catchment area] < 2% -Positive impact of a ratio of [biofilter surface/catchment area] > 2% - <i>Melaleuca ericifolia</i> : best performing species on the long term, probably due to thick roots favouring macropore creation and preferential flows contrary to the dense but fine root system of <i>C. appressa</i>
(Le Coustumer et al. 2007)	-Field survey (infiltration measurement) of 40 biofilters (differing in their age and design) in 3 Australian states (New South Wales, Victoria, and Queensland) -Impact of bioretention design choice in laboratory experiments (125 bioretention columns of 375 mm diameter and 900 mm height, including non-vegetated controls); 14 weeks of trial	5 tested plant species: - <i>Carex appressa</i> - <i>Melaleuca ericifolia</i> - <i>Microlaena stipoides</i> - <i>Dianella revoluta</i> - <i>Leucophyta brownii</i>	
(Le Coustumer et al. 2012)	Evaluation of biofilter design choice on infiltration using laboratory column experiments (75 vegetated and unvegetated columns of 375 mm diameter and 900 mm height); 72 weeks of trial	5 tested plant species: - <i>Carex appressa</i> - <i>Melaleuca ericifolia</i> - <i>Microlaena stipoides</i> - <i>Dianella revoluta</i> - <i>Leucophyta brownii</i>	

Table 1 (continued)

References	Experimental design	Plant species	Main findings
(Muerdter et al. 2016)	Bioretention facility (infiltrating rain garden) of 7 years old, unmanaged for 11 months, with a ratio of [biofilter surface/catchment area] of about 2% (102m ² /4500m ²)	-Initially introduced vegetation: <i>Eutrochium dubium</i>, <i>Solidago rugosa</i> -Spontaneous vegetation (invasive): <i>Polygonum</i> sp., <i>Erigeron</i> sp., <i>Ampelopsis brevipedunculata</i>	-Spontaneous vegetation being the dominant vegetation community assemblage after 11 months of unmanage system: 48.7% of vegetation cover rate by 5 dominant species but 51.3% of vegetation cover rate by 22 spontaneous species -<i>Eutrochium dubium</i>: dominant species (43% of surface cover), with the tallest stems (88.7 cm) and thick roots (up to 2.2 cm diameter); recommended species for introduction into bioretention facilities -<i>Erigeron</i> sp. with deep roots (29.1 cm) may influence on the long-term the bioretention hydraulic functioning
(Selbig and Balster 2010)*	4 experimental rain gardens (ratio 5:1 for [biorefilter surface/catchment area], receiving roof runoff and rainfall), differing in their soil type (sandy versus clay type) and their grass vegetation type (turfgrass versus autochthonous prairie grass)	-Turfgrass: mix of English raygrass, red fescue and Kentucky bluegrass -Mix of autochthonous prairie grass species: <i>Physostegia virginiana</i>, <i>Rudbeckia subtomentosa</i>, <i>Iris pseudacorus</i>, <i>Chelone glabra</i>, <i>Lobelia cardinalis</i>, <i>Scirpus</i> sp., <i>Carex typhina</i>, <i>Pycnanthemum virginianum</i>, <i>Silene stellata</i>, <i>Aquilegia canadensis</i>, <i>Baptisia bracteata</i>, <i>Aster laevis</i>, <i>Monarda</i>, <i>Schizanthium scoparium</i>, <i>Bouteloua curtipendula</i>, <i>Asclepias incarnata</i>, <i>Sagittaria latifolia</i>	-Noticeable performances of rain garden with a ratio of 5:1 to manage 90% of total rainfall precipitation over 5 years -Higher bioinfiltration performances reported for prairie grass than for turfgrass thanks to a deeper root system (143 cm versus 14 cm), a two times more elevated root mass density per soil volume and a cumulated root length superior of 75% in the A horizon, a mean root length superior of 14% in the Bt horizon, and up to 4% increase in soil organic matter associated with an enhanced bioturbation (favoring air and water infiltration)
(Shrestha et al. 2018)	4 bioretention devices (3.72 m ² each): 2 with a high vegetation diversity (7 plant species) and 2 with a low vegetation diversity (2 plant species)	-Low diversity assemblage: <i>Hemerocallis</i> spp., <i>Panicum virgatum</i> -High diversity assemblage: <i>Asclepias tuberosa</i>, <i>Anemone canadensis</i>, <i>Aquilegia canadensis</i>, <i>Symphoricarichum novae-angliae</i>, <i>Baptisia australis</i>, <i>Helenium autumnale</i>, <i>Lobelia cardinalis</i>	-Root mass density slightly higher in the low diversity devices (0.664 mg roots/cm³ of media versus 0.556 mg roots/cm³), probably due to the presence of <i>Panicum virgatum</i> -Peak outflow rates significantly correlated with high diversity devices because of lower root density with shallow root system contrary to <i>Panicum virgatum</i> planted devices where dense root systems would have decreased infiltration; functional diversity is important to consider over species richness
(Virahsawmy et al. 2014)	7 rain gardens varying from 1 to 9.5 years old (Melbourne, Australia), with ratios of [biofilter surface/catchment area] of about 1 to 8%	Mainly monocotyledonous herbaceous plants (<i>Carex</i> sp., <i>Ficinea nodosa</i> , <i>Juncus</i> sp., <i>Poa</i> sp.)	-Higher infiltration rate near plants compared to bare soil, at least 50 mm/h more infiltration (170 mm/h on average) -Non negative influence of <i>Carex</i> species in field testing contrary to results obtained by Lecouster et al. (2012) in bioretention column experiments: probably due to root system maturity in field context/root turnover and preferential flows higher than in laboratory experiments

Table 1 (continued)

References	Experimental design	Plant species	Main findings
(Xia et al. 2018)	1 bioretention device, tested under laboratory conditions, at a scale of 1:1 (1220 mm × 1220 mm × 1220 mm), comprising 2 vegetated parts	-1 herbaceous species: <i>Festuca arundinacea</i> -1 common shrub species: <i>Buxus microphylla</i>	-Infiltration rate and maximum retention capacity slightly more marked in presence of the shrub species (265 mm) than with the herbaceous one (231 mm)
(Yuan et al. 2017)	15 outside rain gardens of 2 m ² surface each ratio 5:1 for [biofilter surface/catchment area], either planted or unplanted	-A mix of prairie grass species (8 perennial and non-graminoid dicotyledonous species and 2 gramineous species): <i>Amsonia tabernaemontana</i> var. <i>salicifolia</i> , <i>Astilbe</i> "Purple Lance," <i>Calamagrostis brachytricha</i> , <i>Filipendula purpurea</i> , <i>Hemerocallis</i> "Golden Chimes," <i>Iris sibirica</i> , <i>Molinia caerulea</i> , <i>Rudbeckia fulgida</i> var. <i>deamii</i> , <i>Sanguisorba tenuifolia</i> "Purpurea," <i>Veronicastrum virginicum</i> -A commercial mix of turfgrass/common grasses (6 species): <i>Agrostis capillaris</i> , <i>Alopecurus pratensis</i> , <i>Anthoxanthum odoratum</i> , <i>Cynosurus cristatus</i> , <i>Deschampsia cespitosa</i> et <i>Festuca rubra</i>	-Higher retention capacity with prairie grass (16.8 mm) than with turfgrass (14.6 mm) -Enhanced retention capacity (5 to 12% more) respectively for prairie grass and turfgrass (after 2 days of antecedent dry period) compared to bare soil -Increased detention time with the prairie grass mix, respectively of 32 and 54%, compared to turfgrass mix and bare soil

*Non-peer-reviewed documents (research report, PhD thesis dissertation)

especially bioretention media composition, could counteract this "naturally occurring" compaction phenomenon. In other green infrastructure devices like tree boxes and trench, engineered "structural soils" involving a mix of stones and soil (sometimes including biochar) have been developed to avoid soil compaction under urban pavement while allowing for tree root penetration and accession to previously infiltrated and stored runoff stormwater (Bassuk et al. 2005). Regarding bioretention cells, it seems that the influence of media structure and composition on hydraulic load-related compaction processes have been understudied, and, obviously, this would also deserve further attention.

Biofilter sizing, design, and media composition

Several studies highlighted the importance of bioretention device size to maintain their targeted hydraulic performances, often leading to the recommendation of a minimum ratio of 2% for [infiltrating biofilter surface/impervious catchment area] to reduce overflow (Le Coustumer et al. 2007, 2009, Hatt et al. 2009a, Selbig and Balster 2010). In fact, of the three Australian biofilters monitored by Hatt et al. (2009a, b), the one scaled to 1% of the impervious catchment area led to more than one-third overflows over 28 monitored rainfall events. These field results were similar with those obtained in greenhouse studies, where a more rapid reduction of media hydraulic conductivity occurred in columns mimicking "undersized" biofilters with an infiltrating surface area of 1% of the impervious catchment area (Le Coustumer et al. 2007). Comparable observations were also obtained with bioretention columns characterized by a substrate layer height of 300 mm vs. 500 mm, the former leading comparatively to more compacted media impeding infiltration (Le Coustumer et al. 2007).

Obviously, the nature and composition of media materials also significantly alter media bulk density and porosity and subsequently overall hydraulic conductivity (recently reviewed in (Skorobogatov et al. 2020)). Amendments of vermiculite or organic matter (in the form of mulch or compost) were hence reported to positively influence water infiltration rates (Le Coustumer et al. 2007, 2012). The addition of organic matter (including in the form of root debris from turnover) is usually well known to result in lower soil compaction and bulk density, favoring deeper air and water penetration (Saxton and Rawls 2006; Skorobogatov et al. 2020). Soils with high silt contents would also tend to favor the growth and development of extensive root systems with large roots (MacDonagh 2015), potentially increasing hydraulic conductivity. A higher median infiltration rate (10.7 cm/h) in sandy-substrate filled rain gardens was reported by Selbig and Balster (2010) over the course of a 5-year field study compared to the 6.35 cm/h rate obtained with clay substrate devices (regardless the introduced vegetation type, turfgrass,

or prairie grass) (Selbig and Balster 2010). Over the 5-year study, the rain gardens monitored by Selbig and Balster (2010) was able to store (and to infiltrate) nearly all the received water due to their 5 to 1 ratio of contributing area to receiving area. Moreover, as reminded by these authors, for stormwater infiltration purposes, the actual retention capacity of a bioretention device should be estimated considering its own water storage volume (due to available media pore space) as well as the underlying soil layer hydraulic conductivity, and inter-event evapotranspiration losses. Thus, in the absence of any underdrain (therefore allowing for direct water exfiltration into the subsoil) and depending on the pedological context, i.e., nature, thickness, and hydraulic properties of the subsurface soil and its specific yields (ratio of water volume drained by gravity in saturated soil/total soil volume (Fetter 2001)), bioretention device surface footprint could be minimized by favoring their vertical implementation and sizing. With this in mind, the lag times for recovering antecedent unsaturated substrate conditions could be easily lowered below guideline values (Selbig and Balster 2010). Moreover, in terms of “urban densification,” the development of more “vertically designed” bioretention systems may offer additional opportunities (depending on local pedological constraints) for extensive implementation of these green infrastructures at the city scale. If infiltration is the main objective, however, more attention should be also given to laterally infiltrating water fluxes (i.e., enhancing vertical infiltration rather than infiltration through the sidewalls) to avoid side effects towards building foundations. In such contexts, the influence of vegetation, either initially introduced or spontaneous species, may reveal as a determining factor to maintain the bioretention media hydraulic properties in accordance with the initially targeted hydrological performances, especially concerning infiltration rates.

Evaluation of the influence of plant functional type and traits quantitative stormwater management

General field study considerations regarding runoff treatment with vegetative buffer strips

Grass barriers as well as vegetative filter strips (> 5 m width) and hedge systems have all been reported to be effective in reducing runoff (Muñoz-Carpena et al. 1993; Gilker et al. 2002; Blanco-Canqui et al. 2004a, 2004b; Rachman et al. 2004; Seobi et al. 2005; Sharma et al. 2006; Udawatta et al. 2006). A brief reviewing of the general findings about their runoff treatment capacity may help understanding the influences that different plant types may have also in bioretention systems. Early, numerical modeling approaches simulating the hydrology of a vegetative filter strip (and involving a modified infiltration submodel based on the Green-Ampt equation) indicated that runoff volume as well as velocity

and peak discharge was lower for soils with denser grass cover and higher infiltration rates (Muñoz-Carpena et al. 1993). Later on, field studies conducted by Blanco-Canqui et al. (2004a) showed that traditional fescue (*Festuca arundinacea*) filter strips and more particularly when combined with switchgrass (*Panicum virgatum*) barriers were able to significantly reduce runoff compared to a continuously cultivated fallow treatment only (which also acted as a source of runoff-conveyed sediments). Indeed, an additional barrier of switchgrass of 0.7 m × 1.5 m above a fescue filter strip of 7.3 m × 1.5 m allowed for a 15% of runoff reduction compared to 11% for a fescue filter strip only of 8 m × 1.5 m (Blanco-Canqui et al. 2004a). According to the authors, this would have been probably due to increased runoff detention time in relation with dense surface debris as well as deep rooting system in the switchgrass barrier improving infiltration (Blanco-Canqui et al. 2004a). Similar results using switchgrass barrier were previously reported by Gilley et al. (2000). Besides, the use of a vegetative filter strip barrier made of a mixture of native species (Eastern gamagrass (*Tripsicum dactyloide*), Indian grass (*Sorghastrum scoparium*), big bluestem (*Andropogon gerardii*), gray-head coneflower [*Ratibida pinnata* (Vent.) Barnhart], and purple coneflower [*Echinacea purpurea* (L.) Moench]) was as effective as the fescue filter strip alone for runoff reduction, which could have been explained also probably by the development of a sufficiently dense and deep root network by the native species mixture favoring infiltration (Blanco-Canqui et al. 2004a). Nevertheless, as pointed out by Blanco-Canqui et al. (2002), subsequent water infiltration and percolation into deeper and slower permeable Bt horizon soil would dominate in the long-term. Hence, switchgrass barriers would be effective for reducing runoff by enhancing infiltration in fine-textured (e.g., claypan) soils (Blanco-Canqui et al. 2004a). Further, field studies confirmed the efficiency of both dormant and active stiff-stemmed grass like switchgrass to disperse and temporarily pond concentrated runoff as well, presumably involving better infiltration while resulting in substantial increases in sediment and nutrient trapping compared to fescue filter strips alone (Blanco-Canqui et al. 2004b). Rachman et al. (2004) measured the field-saturated hydraulic conductivity of stiff-stemmed ten years old grass hedge systems (*Panicum virgatum* L.). They confirmed their positive influence on infiltration rates in (fine) silt loam soil, with more than seven times and twenty-four times greater saturated hydraulic conductivity respectively, within the grass hedge than within a row crop zone or a deposition position (upslope from the hedge). Grass species (e.g., Eastern gamagrass [*Tripsicum dactyloides* (L.) L.] may be particularly tolerant to adverse soil conditions such as acidity, Al-toxicity, and high soil strength conditions, rendering them of great interest for the establishment of vegetative conservation barriers in harsh environments (e.g., dense or acid soil)

(Gilker et al. 2002), therefore, explaining their efficiency to improve soil hydraulic properties in such context (Blanco-Canqui et al. 2004a, 2004b). Saturated hydraulic conductivity (K_{sat}) in claypan horizon, water retention, pore-size distributions, and bulk density were all reported to be positively impacted by a grass-legume (redtop (*Agrostis gigantea* Roth), brome (*Bromus* spp.), and birdsfoot trefoil (*Lotus corniculatus* L.)), and agroforestry (pin oak (*Quercus palustris* Muenchh.), swamp white oak (*Q. bicolor* Willd.), and bur oak (*Q. macrocarpa* Michx.)) buffer strip compared to a row crop treatment (corn—*Zea mays* L.—and soybean—*Glycine max* (L.) Merr.) (Seobi et al. 2005). Indeed, 2.3% lower bulk density, 3 to 33% higher coarse mesoporosity (60 to 1000 μm pore diameter), and 3 to 14 times higher hydraulic conductivity were reported in grass and agroforestry treatment respectively, compared to the row crop treatment (corn (*Zea mays* L.) and soybean [*Glycine max* (L.) Merr.]) (Seobi et al. 2005). These resulted also into substantial increases of water storage potential (on average 1 cm per 30 cm depth) (Seobi et al. 2005), highlighting the effectiveness of vegetative filter strips and hedges on runoff reduction. Computed tomography analyses of soil pore parameters was predictive of K_{sat} , clearly establishing correlations between K_{sat} and macro-/meso-porosity, as well as number of pores (Udawatta et al. 2006). Overall, largest effects on pore parameters were obtained under tree cover, followed by grass and row crops (Udawatta et al. 2006). Hence, based on vegetation information, Sharma et al. (2006) were able to develop pedotransfer functions (relying on machine-learning algorithms) at fine scale for estimating soil hydraulic properties including K_{sat} . From all of these works on vegetative buffer strips, intuitively similar observations could be reported for the positive role of grasses as well as shrubs on runoff mitigation and infiltration enhancement in bioretention systems, especially in the presence of clay-type media.

Lawn and prairie grass influences on stormwater infiltration in bioretention systems

Infiltration measurements from in situ rain gardens usually indicated that regardless of substrate type, devices planted with prairie grasses evidenced greater infiltration rates than those planted with turf (domestic) grass, highlighting the importance of root functional diversity on vertical water fluxes. For instance, Selbig and Balster (2010) reported seasonal effects among herbaceous vegetation with enhanced infiltration rates in spring compared to summer for prairie grass, whereas no clear distinction could be made for turfgrass. Moreover, these authors showed that vegetation type interactions with rain garden substratum were more evident in clay rich soil than in sandy media, leading to more than a threefold increase in infiltration rate, i.e., from 0.63 to 2.24 cm/h (Selbig and Balster 2010). Overall, infiltration

rate increased steadily over the 5-year monitoring study for all planted devices, still with higher hydraulic conductivity measured in the presence of prairie grass (Selbig and Balster 2010).

Root profile analyses (conducted on whole soil profile up to a depth of ~ 1.5 m) usually revealed much contrasted root distributions according to the type of grass cover (prairie vs. turfgrass) and to the soil depth. While many fine roots of the turfgrass had colonized the first 14 cm (up to 17 cm) of rain garden in the 5-year study of Selbig and Balster (2010), they reached a higher depth of 143 cm for prairie grasses, with many roots even observed in the Btg horizon (i.e., “deep” horizon containing illuvial clays). It should be noticed that the deeper root colonization under this kind of “wild type” plant cover was generally also associated with greater macrofaunal (earthworm) activity, further promoting pedobioturbation processes with consequently deeper air and water movements into the substrate of the clay-rich rain gardens (Selbig and Balster 2010). Moreover, the higher organic matter contents measured in the A-horizon planted with prairie grasses (+ 4% compared to that of the lawn grasses) also correlated with a lower soil density (Selbig and Balster 2010), a phenomenon likely to enhance water infiltration (Skorobogatov et al. 2020). Regarding root morphology and functional traits, the mean root mass density per soil volume (RMD) of prairie grass in clay-type rain garden in this study was practically twice that of turfgrass in the A horizon superficial soil (Selbig and Balster 2010). The mean specific root length (SRL, i.e., root length per unit of root dry mass) was 23% and 14% higher respectively in the A and Bt horizons of prairie grasses compared to turfgrasses (Selbig and Balster 2010). Such observations indicated that roots of prairie grass were overall more numerous and thinner than those of turfgrass (both in the A and Bt horizons), allowing for more efficient and deeper soil exploration. Besides, the cumulative root length density (RLD, root length per unit volume of soil) was 75% greater in the A horizon of prairie grasses than turfgrasses (Selbig and Balster 2010). Plant species demonstrating high cumulative RLD and SRL may be more competitive for a rapid exploitation of soil water resources at a lower biological cost (i.e., energy, carbon cost) at the individual level. For instance, Eissenstat (1991) reported a faster water use for orange trees with such root morphology than for those with shorter SRL. These different root trait values between the two types of grasses could here also explain the increased ability of prairie vegetation to grow roots to significantly deeper soil depths in clayed substrates (Selbig and Balster 2010). Similar modeling results have been more recently reported by (Bloorchian et al. 2016), confirming the higher efficiency of prairie grasses in a grass strip context to retain and to infiltrate runoff compared to traditional lawn grasses. According to their work,

the extent to which grassy plants could promote infiltration would positively interact with the type of soil and its initial hydraulic conductivity, with in order of increased runoff reduction: silty clay < silt loam < loamy sandy, as well as the type of plant cover (i.e., enhanced infiltration with prairie grass compared to turfgrass) (Bloorchian et al. 2016). In fact, because the matrix flow in clay is the least, root-induced macropores will have the greatest effect in this kind of media. As a result, it has been shown that prairie grasses were likely to reduce runoff by 37% at a silty-clay soil, compared to 10% abatement for lawn grasses (Bloorchian et al. 2016). It should be noticed that such observations have been thoroughly documented in the literature and do not relate specifically to bioretention media (see the section before and references therein). Yuan et al. (2017) also demonstrated greater retention and detention capacities in bioretention devices planted with a taxonomically diverse community composed of mixed forb-rich herbaceous perennials and two grass species compared to a commercial mixture of six typical law land grass species (Table 1). The rich mixed perennial herbaceous community thus evidenced an additional 2.2 mm of precipitation retention compared to law land grasses, with respectively 16.8 and 14.6 mm of retention for each cover type (Yuan et al. 2017). Obviously, in this study, the length of antecedent dry period as well as the interaction of this period with the type of vegetation cover was found to be significant in terms of (positive) influence on the media retention capacity of the corresponding devices (i.e., lesser moisture meaning more retention). Approximately 12 to 5% additional water retention capacity could be estimated in the rich perennial and lawn grass mixtures respectively compared to bare soil (after 2 days without precipitation) (Yuan et al. 2017). Regarding the detention capacity, the rich perennial mix also increased detention time by 32 and 54% before observing peak runoff (compared to the lawn grass mix and bare soil respectively) (Yuan et al. 2017). Consequently, runoff intensity was decreased at an average rate of 0.4 mm/min for the rich perennial-dominated mixture compared to bare soil for a simulated extreme rainfall intensity of 72.6 mm/h (Yuan et al. 2017).

Taken together, all of these works from various experimental design and geographical origins eventually suggested favoring “wild type” plants as well as fostering plant taxonomic community diversity (and in fact concomitant functional diversity) in order to enhance (bio)infiltration performances. Indeed, results obtained by Shrestha et al. (2018) indicated a positive correlation between peak discharge, precipitation volumes, and high plant species richness. Such observations would have been due to the plant selection criteria in their study, which was done only on a taxonomic basis instead of functional traits. The root biomass measurements (i.e., root biomass density expressed in

mg/cm³ of soil) revealed a slightly higher root density in the “low diversity” experimental units planted with *Panicum* sp. (i.e., 0.664 vs. 0.556 mg roots/cm³ substrate). According to the authors, this gramineous species, which is known for its deep and dense root system would have led to sufficiently deep and dense root proliferation in the “low diversity treatment” compared to the “high diversity treatment,” in order to slow infiltration in the sandy bioretention media (Shrestha et al. 2018). In fact, such results would echo those from (Lu et al. 2020) who reported that fine roots at high density could decrease macropores while increasing pore volume and thus reducing saturated hydraulic conductivity (*K*_s) contrary to water retention capacity, especially in coarse-textured soils. By contrast, as reported by Lu et al. (2020) for fine-textured soils, positive root influences would be expected for *K*_s, as it was also the case in the study of Selbig and Balster (2010) for clay-type rain garden.

Sedge and shrubs effects

In their former study regarding the bioretention design choices impacting hydraulic performances, Le Coustumer et al. (2007) included the comparison of five common biofilter plant species differing from lawn grasses (*Carex appressa*, *Dianella revoluta*, *Microlaena stipoides*, *Leucophyta brownii*, *Melaleuca ericifolia*), among which *C. appressa* proved to be significantly superior to the others to promote infiltration (Le Coustumer et al. 2007). According to the authors, the overall trend observed with this plant species at this course of the study could have resulted from its dense vegetative system, contrary to more sparse *M. ericifolia* cover and shallow root system of *M. stipoides* (or simply the absence of vegetation) (Le Coustumer et al. 2007). However, *M. ericifolia* revealed as the best performing species a few months later, with a decrease in hydraulic conductivity from 155 to 100 mm/h during the first 40 weeks followed by an increase to 295 mm/h after 60 weeks, to finally reach a value of 245 mm/h at 72 weeks (Le Coustumer et al. 2012). On the contrary, columns planted with the other species including *C. appressa* were associated with a decrease in media hydraulic conductivity until reaching an asymptote similarly to the unplanted columns after 72 weeks of testing (Le Coustumer et al. 2012). Here again, a comparative analysis of root architecture (length, specific surface area, and diameter) and profile (depth distribution) between *M. ericifolia* and *C. appressa* enabled to distinguish two contrasted morphologies. RLD of *C. appressa* was found to be approximately ten times greater than that of *M. ericifolia* down to a depth of 63 cm where the two species met (Le Coustumer et al. 2012). However, the average diameter (d₅₀) of *M. ericifolia* roots (0.09–0.25 mm) was about 1.5 to 2 times larger than those of *C. appressa* (0.08–0.15 mm), meaning that the dense but thin root network of *C. appressa*

was less favorable for water infiltration than the thicker roots of *M. ericifolia* (Le Coustumer et al. 2012). Further experimental demonstration was made showing that *M. ericifolia* was able to maintain and even increase substrate hydraulic conductivity with infiltration rates measured at 312 mm/h, i.e., twice the initial value of 155 mm/h (Le Coustumer et al. 2012). The thicker roots of *M. ericifolia* would have therefore enhanced macropore formation whereas the thinner and denser roots of *C. appressa* would have been less favorable to water infiltration and percolation (Le Coustumer et al. 2012). Despite the fact that pollutant removal is out of the scope of the present review paper, enhanced hydraulic conductivity raises the concern that macropores may be bypassing the media matrix. In such cases and depending on the implementation context of bioretention systems, special attention should be paid to what extent it may compromise the removal performances of media stressors. More recently, soil column experiments (out of the context of bioretention) reported by Boldrin et al. (2022) tended to confirm the positive influence of thicker branched root system, in this case from the legume *Lotus corniculatus* on saturated hydraulic conductivity ($3.9\text{e}^{-5} \pm 1.2\text{e}^{-6} \text{ m.s}^{-1}$) compared to the grass *Festuca ovina* ($9.9\text{e}^{-6} \pm 1.3\text{e}^{-6} \text{ m.s}^{-1}$). It is important to point out that among the five plant species previously tested in the long-term study (72 weeks) of Le Coustumer et al. (2012), none of them had stopped water infiltration. Finally, the hydrological performance of a laboratory-scale bioretention device planted with a variety of “*Buxus sinica*” shrub compared to tall fescue also demonstrated a higher infiltration rate in the presence of the shrub species (for inflow rates of 6 mm/h) (Xia et al. 2018), presumably due to its thicker root system. For an inflow volume representing a rainfall precipitation of 1220 mm, the maximum storage capacities were of 265 mm and 231 mm for *Buxus sinica* and tall fescue, respectively (Xia et al. 2018). In fact, it is well known that trees and shrubs in slowly infiltrating soils usually concentrate flows near the trunk and into the soil there due to coarse macropores. Thus, such phenomenon could have been also illustrated here with *Buxus sinica*. During this experiment, it was also worth to notice that the outflow peaks could be delayed by at least 13 min (depending on inflow rates) and reduced by at least 52%, reinforcing the potential role of vegetated biofilters in runoff mitigation and stormwater retention (Xia et al. 2018).

Case of hybrid stormwater-greywater bioretention devices

In order to alleviate the plant physiological stresses inherent to prolonged dry periods between rainfall events, including severe summer drought, biofilter irrigation by greywater has recently retained attention (Barron et al. 2019, 2020). Within this context, results described in Barron et al. (2019, 2020) from biofilter columns aged of 20 months were in line with

those previously reported by Le Coustumer et al. (2012) regarding the vegetation effects on stormwater runoff infiltration rates. Indeed, a higher infiltration rate (here determined as the amount of water outflow from the bottom of 94-cm columns over time) was reported in the early phases of the two studies of Barron and colleagues (Barron et al. 2019, 2020) in the presence of plant species with thick, rhizomatous roots (e.g., *Canna x generalis*) or fleshy roots (e.g., *Phormium tenax*). On the contrary, denser and finer root plants like *Carex appressa* appeared to rapidly impede water percolation (Barron et al. 2019, 2020). Between these two contrasting root morphologies, the root of *Phormium tenax* that was sparsely “branched” and fleshy, with steady aboveground and belowground biomass production, was related to an average hydraulic conductivity that remained the most constant over the course of the two successive studies (Barron et al. 2019, 2020). Alternating dry and wet periods also substantially improved infiltration rates regardless of the presence of plants, probably due to biofilm desiccation and pore development including macropore formation following surface media “cracking” (Barron et al. 2020). Moreover, suction head was also undoubtedly higher within such contexts, which would have also increased initial infiltration rates. However, as mentioned by the authors, 3-year experiments showed an overall decrease in hydraulic conductivity presumably due to the physical obstruction of media pore space by constrained large root volumes in greenhouse bioretention columns (Barron et al. 2020). To our knowledge, no such root-induced media pore obstruction has been reported for bioretention field studies in the literature. Instead, field assessment of soil hydraulic conductivity in a range of biofiltration systems (rain gardens, (infiltrating) bioretention ponds) usually confirmed a seasonality effect with higher infiltration rate values in spring–summer than in autumn–winter. This could be related to the replenishment of the bioretention media water storage capacity in dry seasons as well as to the vegetation growth dynamics associated with bioturbation phenomena during these intense periods of soil biological activity (Selbig and Balster 2010; Gonzalez-Merchan et al. 2014; Muerdter et al. 2016; Hart 2017). Moreover, this could be also reinforced by the fact that water viscosity decreases as temperature increases, resulting in enhanced soil hydraulic conductivity as reported by (Emerson and Traver 2008). Furthermore, increased microbial activity in situ and more generally temperature were also likely to alter water infiltration rates following organic matter mineralization (including plant materials) and macropore formation. This latter could be enhanced by (micro)-cracks in the surface media at the hottest periods (Nogaro and Mermillod-Blondin 2009; Gonzalez-Merchan et al. 2014), resulting in increased infiltration and percolation once raining.

Other field works also reported the significant influence of vegetation beyond the media surface, involving the formation and development of preferential water fluxes from

aerial parts to the roots (stemflows), particularly in the case of tall grasses (to some extent similar to shrubs and trees). Moreover, wind-induced movement of the stems or rigid foliage of various species belonging to grasses and shrubs can disrupt the clogged sediment surface layer (at the base of plants), creating macropores that channel water directly to the root systems (Martinez-Meza and Whitford 1996). Virahsawmy et al. (2014) showed that average infiltration rates (K_{sat}) measured around species like *Juncus* sp., *Carex* sp., *Ficinia nodosa*, and *Poa* sp. (in seven bioretention systems located in Melbourne, Australia) were significantly higher than in unplanted areas. In this study, an additional mean infiltration rate of 50 mm/h, reaching up to 170 mm/h in fine sand substrates, had been measured near plants regardless of the clogged state of the surface devices (Virahsawmy et al. 2014). Interestingly, contrary to previous greenhouse experiments where *Carex appressa* species did not increase infiltration (Le Coustumer et al. 2012), results obtained by Virahsawmy et al. (2014) in full-size bioretention devices clearly suggested a role of root senescence and turnover, given the age of the infrastructures, that would lead to enhanced infiltration in the long term for vegetated soils. In other words, bioretention column experiments (such as in Le Coustumer et al. (2012)) may not have been conducted for a sufficiently long time to allow for natural (fine) root senescence and turnover leading to higher soil porosity and concomitant preferential flows over time (Virahsawmy et al. 2014), contrary to what has been reported in natural ecosystems (Archer et al. 2002). Moreover, in column experiments, root development are constrained by the column geometry and finite soil volume, which may result in soil pore blocking (and infiltration impediment) even in long-lasting experiments (e.g., 3 years) (Barron et al. 2020). At the field scale, where roots are less space constrained by the bioretention cell design, root (and stem)-induced preferential flows and enhanced infiltration would be generally favored under various degrees of surface clogging, as reported by Virahsawmy et al. (2014). Another hypothesis regarding the lack of root-induced infiltration in some laboratory/column studies may be related to the fact that column experiments may tend to be more heavily loaded than comparable field systems. Despite the fact that artificial stormwater volume addition usually try to mimic what has been observed in the corresponding locality in terms of rain intensity, rain episodes are not always occurring on a so much regular basis (e.g., every weeks or two times per week for instance) as it is sometimes considered in laboratory experiments. However, antecedent dry period can affect evapotranspiration processes (and subsequent replenishment of water storage volume) as well as overall root development and rhizosphere-associated processes (including those microbially mediated) involved into root and organic matter turnover as well as soil porosity modifications. To which extent, all of these processes

may affect resulting bioretention media hydraulic properties, depending on artificial stormwater regime in column experiments, would deserve further investigations. More systematic in situ assessments of infiltration rates at plant scale/near plants would also help to better understand (and model) the effects of plants and their associated root system architecture in the modifications of bioretention media hydraulic properties.

Tree seedlings

As recently reported by Xie et al. (2020) in a study beyond the scope of the present review focusing on bioretention, the presence of trees with a deep root network could increase initial soil infiltration of cities by 53–330% compared to grass cover type. More closely to bioretention systems, Bartens et al. (2008) (reprinted in (Day et al. 2008)) investigated the ability of three tree species to grow in a silty-clay soil, focusing on their respective root development characteristics (taproot vs. fascicular root architecture) into compacted soils (1.3 or 1.6 g/cm³) similar to those potentially encountered under impermeable pavements near bioretention systems. Two-year-old seedlings of black oak (*Quercus velutina* Lam.) and red maple (*Acer rubrum* L.) confirmed their ability to grow well in such media, with a normal root development regardless of soil compaction level (Bartens et al. 2008). Although soil hydraulic conductivity (K_{sat}) decreased over a growing season for all the experimental conditions, the presence of seedlings allowed the maintenance of a water infiltration rate on average 63% higher in planted containers than in bare soils (Bartens et al. 2008). Infiltration rates even reached up to 153% when considering only the vegetated soil condition of the highest compaction level (1.6 g/cm³) vs. unplanted one (Bartens et al. 2008). Moreover, water drainage generally occurred faster when planted with oaks compared to maples, probably due to the coarser root system of the former (Bartens et al. 2008). For red ash (*Fraxinus pennsylvanica* Marsh), which was the third tree species monitored under outdoor growth conditions for 2 years, hydraulic conductivity measurements also showed much faster drainage in highly compacted soils (1.6 g/cm³), with an average infiltration rate 27 times greater than in bare soils (Bartens et al. 2008). Again, these observations highlighted the importance of species selection according to their root traits (e.g., root thickness) in order to enhance infiltration processes related to the development of preferential flows following deep root penetration into the substrates (Bartens et al. 2008). However, while this should have been particularly relevant for thick rooted species (Bartens et al. 2008), their influence on water lateral flow still needs to be investigated. Moreover, taking into account that these studies were on field studies, i.e., not related to thinner and more permeable bioretention media, further investigations

would be needed for better characterizing tree root influences upon infiltration in more related bioretention systems, such as for instance “stormwater tree boxes” that were recently described by Seidl (2019). Besides, despite the fact that bioswale or infiltrating tree trench systems are green infrastructures which may be considered out of the scope of this review dealing on “bioretention” *stricto sensu*, it should be worth noting that oversizing such systems may result in water deficit altering tree growth (Scharenbroch et al. 2016; Tu et al. 2020). If a tree is intended to be introduced into a rain garden-like system for instance, careful consideration should be also given to its water requirement to avoid any hydric stress and to provide sufficient soil depth for perennial growth (personal communication).

Quantitative assessment of the relationships between root functional traits and the increase of media hydraulic conductivity

Experimental reports

While there should be no ambiguity on the existence of plant and more particularly root trait influences on water fluxes and especially infiltration in bioretention devices, only very few studies to date have focused on the elucidation of the quantitative relationships between these biological characteristics (including potential seasonality effects) and water infiltration. This has been recently investigated by Hart (2017) through field measurements of infiltration (estimated through water content variation according to substrate depth) in parallel with thorough root trait evaluation in two categories of biofilters. These contrasted in their vegetation cover type: one mainly dominated by common rushes (*Juncus patens*) and the other by the *Carex* genus (mainly *C. estacea* and *C. dolichostachya*). Overall, *J. patens* revealed as a better performing species than *Carex* sp., with positive correlations when plotting infiltration against root traits (Hart 2017). Only root surface area density (RSAD; total root surface area per unit of soil volume) determined in the surface layers showed a positive relationship with infiltration in the presence of *Carex* sp. during the growing season (Hart 2017). In early summer (May–June), *J. patens* root mass density (RMD) and root diameter (RD) in the upper media layer (0–51 cm) explained 77% of the variability of the measured infiltration rates (Hart 2017), emphasizing root trait influences as well as seasonality effects on the mediation of infiltration processes from the biofilter surface horizons. From a biological perspective, RMD and RD reflect plant investment in root growth and maintenance processes usually associated with an increase in the other related root morphology parameters (e.g., root surface area density—RSAD; root volume density (total root volume per unit of soil volume)—RVD). Such root development and growth (i.e., gains in thickness

and length) is likely to enlarge “rhizopores” (spaces between the soil and the root surface) due to the compression of soil aggregates during root penetration, these macropore developments being further enhanced by root turnover or senescence processes (Archer et al. 2002; Hart 2017; Mair et al. 2021). These root patterns were nonetheless generally more marked in the deeper bioretention media and subsoil layers (51–100 cm depth) than in the surface layer of the studied green infrastructures, further associated with positive correlations with the estimated infiltration rates (Hart 2017). The high root growth (+ 66% RMD) observed in the late summer during this study also echoed previous observations reported by (Gill and Jackson 2000) for wetland plant species (with an overall root turnover of 55%) and for plants of other geographically distinct rain gardens and bioretention ponds (Hatt et al. 2009a; Selbig and Balster 2010; Gonzalez-Merchan et al. 2014). Furthermore, according to Hart (2017), the dry summer conditions could have favored also the formation of macropores with enhanced aeration of deep media layers, a phenomenon formerly suggested by Selbig and Balster (2010) in similar field evaluation context. Naturally occurring root shrinkages with diurnal variations, sometimes reaching up to 60% of the maximum diameter for fine roots (Huck et al. 1970), may eventually increase the enlargement of previously formed macropores and thus the establishment of preferential flows along the roots with potentially enhanced infiltration during rainy episodes following a drought period (Archer et al. 2002; Hart 2017). These processes in conjunction with increased temperatures could also explain the seasonal higher root growth investment often reported from field survey in an effort of roots to access deeper moist layers during drought periods (Mitchell et al. 1995; Hart 2017).

Still considering plant trait evaluation in relation with infiltration rate, the RMD values reported by Hart (2017) in his work, ranging from 0.1 to 2.2 kg root/m³, greatly exceeded those previously reported by Bartens et al. (2008), ranging from 0.1 to 0.7 kg/m³ (also related to a positive root impact on hydraulic conductivity in similar silty-clay soil type). Moreover, for root density (RD) and more specifically the ratio between “surface soil” RD and “subsoil” RD, the measured values between 1 and 1.6 by Hart (2017) were in line with those reported by Zedler (2007), i.e., between 1 and 2 for wetland vegetation, which would be likely to positively impact infiltration in bioretention devices. Besides, according to the author, a minimum root diameter of 1 to 1.2 mm could be in fact required to promote macropore formation from the natural processes of root growth, exudation and media compaction, shrinkage, and senescence. Importantly, this work suggested that little variations in root traits, especially the average root diameter in the upper media layers (i.e., varying from 0.9 mm (*Carex* sp.) to 1.2 mm (*J. patens*)),

were likely to induce major consequences on infiltration rate (Hart 2017), thus reinforcing the importance of plant species-related functional trait selection. For the best performing species *J. patens*, a root diameter increase of 0.2 mm (i.e., varying from 1.1 to 1.3 mm) would lead to an infiltration enhancement from 2 to 10 cm/h (Hart 2017).

More recently, the work of Johnston et al. (2020) confirmed the non-neglected influence of root depth distribution on the modification of bioretention media porosity, especially the impact of macropore distribution at a mesocosm-scale experiment. These latter seemed to be more connected, thereby promoting drainage under prairie grass and shrub cover contrary to domestic lawn type grass (and bare soil) (Johnston et al. 2020), and potentially bypass flow (with lower media stressor removal). For small stormwater inputs (8–50 mm during 1 day of stormwater application), the prairie treatment had a shorter lag-time than the control treatment as well as the least peak flow rate (6 mm.h⁻¹), which was also lower than the one of turfgrass (15 mm.h⁻¹). A marked influence was observed overall on the reduction of drainage water volume within vegetated devices regardless the plant functional type, because of the alteration of media porosity (structure and particularly the assumed pore connectivity) leading to improved water storage available for evapotranspiration (Johnston et al. 2020). However, a greater volume of drained water was measured for moderate rainfall events (<90 mm) for turfgrass compared to prairie grass (likely related to a different pore structure in deeper substrate layers due to lower root density) and pointed out hence the complexity of vegetation cover effects on bioretention hydrological performances in parallel with system maturation (Johnston et al. 2020).

As recently reported following field assessment, the hydraulic conductivity of many mature bioretention systems could be maintained several years after construction (Sprakman et al. 2020; Sprakman and Drake 2021). Some of them drained very rapidly (~no ponding zone even under extreme flow), preventing an adequate residence time for dissolved contaminant removal (e.g., Rodgers et al. (2022)). In fact, designer concerns about clogging may sometimes lead to the selection of bioretention media possessing hydraulic conductivity higher than necessary. Within this context, we argue that trade-offs between finer-textured media (improving residence time and soluble pollutant sorption) and more appropriate selection of plant species/root functional traits, to positively influence infiltration (and potentially rhizosphere-mediated pollutant dissipation), may help to design more effective biofilters regarding both qualitative and quantitative stormwater treatment.

Numerical modeling approaches

Recently, Kumar and Singh (2021) investigated the influence of plant density on infiltration rates using deep learning techniques and gradient boosting machine (under outdoor

mesocosm experiments). Both machine-learning methods performed far better than the popular Philip's infiltration model (Philip 1957) with almost 99% of the predicted values in both cases which were closed to the perfect agreement line with measured field data (compared to 75% agreement or less with Philip's model) (Kumar and Singh 2021). Gradient boosting machine performed slightly better than deep learning algorithms with lower computational resources required. Accordingly, results showed that plant density was significantly enhancing water infiltration rate (Kumar and Singh 2021). Despite the fact that using this kind of soft computing methods did not allow to choose for the most efficient plant type according to the experimental design (Kumar and Singh 2021), the accuracy of the obtained predictions highlighted the usefulness of such innovative modeling approaches, paving the way for further experiments involving additional testing of other robust classification methods (e.g., random forest classification). By this way, one could hope to achieve a better evaluation of plant functional type effects on infiltration, relying on a restricted number of experimental data for machine-learning training and regardless a thorough knowledge of plant biophysics. Consequently, designers and practitioners could suggest more appropriate plant selections given local environmental conditions and targeted hydrological performances.

However, in order to better understand and predict the influence of plant functional types and traits on stormwater infiltration rate, there is no doubt that a biophysical model directly linking root biological attributes to water fluxes in soil would be a highly desirable tool. To our knowledge, nevertheless, a thorough biophysical modeling of water infiltration rate taking into account root architecture as well as its three-dimensional (3D) distribution in bioretention media is still lacking. Future works should take benefits of the modeling capability offered by the coupling of three-dimensional root architecture models with 3D subsurface flow models. Recently, for instance, Mair et al. (2021) revisited mathematical models of moisture transport for predicting root-induced preferential flows in vegetated soils. By this way, they were able to demonstrate that different preferential soil moisture transport occurred along root axes of willow and grass respectively, in relation with soil pore-water pressure profiles (corresponding to saturated hydraulic conductivity values) varying from one plant species (i.e., root architecture) to another (Mair et al. 2021). According to our literature review, this is probably one of the first work to date to explicitly model the effects of root architecture *stricto sensu* on water infiltration. Extended to the vegetation palette used in bioretention systems, such model should strongly improve our understanding of the influence of plant functional traits in bioinfiltration performances, also offering new opportunities for better plant choice adapted to local constraints and long-term effectiveness prediction.

Considerations regarding spontaneous versus introduced plant species on bioinfiltration performances

Design of bioretention systems including successive media and drainage layers of varying water retention capacity offers fluctuating environmental conditions and additional constraints to the perennial growth and development of the introduced plant species (e.g., cyclical flooding, extended period of substrate drying). Spontaneous vegetation sometimes proves to be better adapted to the local media hydrological regime, its compaction level as well as the overall quality of receiving urban runoff (i.e., nutrient and micropollutant concentration and loads). Therefore, in unmanaged biofilters, weeds and other unwanted species may grow invasively, often dominating the local plant communities (Hatt et al. 2009a; Muerdter et al. 2016). As an illustration, floristic surveys conducted by Muerdter et al. (2016) in a 7-year-old biofilter showed the importance of vegetation maintenance when the perennial establishment of the introduced plants is intended. Indeed, flora species counts in this work indicated that spontaneously colonizing specimens represented more than half of the vegetated surface (51.3% coverage by 22 plant species) following 11 months of biofilter non-maintenance, as opposed to the coverage by initially voluntarily introduced species (48.7% coverage by 5 plant species) (Muerdter et al. 2016). Moreover, among the six identified dominant species in this study (*Eutrochium dubium*, *Polygonum* sp., *Erigeron* sp., *Ampelopsis brevipedunculata*, *Solidago rugosa*, and a stiltgrass species), only two of them, i.e., *Eutrochium dubium* (43% coverage of the vegetated area) and *Solidago rugosa*, had been voluntarily introduced (Muerdter et al. 2016). Interestingly, the evaluation of root functional traits showed that *E. dubium* possessed the thickest roots (up to 2.2 cm in diameter) associated with a dense and extensive aboveground biomass (also with the tallest stems, 88.7 cm on average) within the plant community (Muerdter et al. 2016). However, the average longest roots were similar among the three species dominating this plant community (*E. dubium* and *S. rugosa*—introduced species; *Erigeron* sp.—spontaneous species), reaching approximately 29.1 cm (Muerdter et al. 2016) and meaning that they had probably not extended all the way through the media at this time of sampling. Unfortunately, no link could be established during this field survey between root traits and local infiltration rate due to a lack of correspondence between the different vegetation and infiltration measurement quadrats. Nevertheless, considering the reported survival and dominance of *E. dubium* characterized by a dense surface cover percentage and a thick root network, this species seemed to be particularly adapted to the studied biofilters and could be recommended for such greening purposes in this temperate climate region (Mid-Atlantic, Maryland, USA). Moreover,

the rapid biofilter colonization by *Erigeron* sp. (spontaneous species) also possessing a deep root network that can reach deeper humid soil layer and therefore influence the evapotranspiration water fluxes clearly raised questions about its beneficial presence for bioretention system. Since this species does not require any specific maintenance (Muerdter et al. 2016), it could be hypothesized that its positive influence on the long term would become decisive to warrant adequate biofilter hydrological functioning.

Positive impacts of spontaneous flora on soil hydraulic conductivity were also confirmed by Gonzalez-Merchan et al. (2014) while monitoring an infiltration basin near Lyon, France (temperate climate), colonized by *Phalaris arundinacea*, *Polygonum mite*, and *Rumex crispus*. Indeed, infiltration rates (determined as saturated soil hydraulic conductivity— K_{sat}) were higher in the vegetated areas compared to non-vegetated ones (Gonzalez-Merchan et al. 2014). Moreover, some relationships could be drawn between plant species, their morphology, and measured infiltration rates. The best performing species was *Phalaris arundinacea*, with a significant increase (by a factor of 2) in the median hydraulic conductivity value in the plant vicinity (Gonzalez-Merchan et al. 2014). Although *P. arundinacea* roots were thinner than those of *Rumex crispus* (the least efficient species shown to promote infiltration in that bioretention pond), its mode of propagation by cuttings (including aerial roots) and rhizomes would have allowed for the maintenance of a greater media porosity and permeability compared to the other species (and to unvegetated areas) (Gonzalez-Merchan et al. 2014). However, it must be pointed out that *Phalaris arundinacea*, *Polygonum mite*, and *Rumex crispus* can be considered as quite noxious and invasive species, and therefore, other species should be preferred when possible.

Although the mechanisms involved into hydraulic conductivity increases is often not clearly elucidated in situ (i.e., preferential channels formed by stem/leaf movements or influence of roots), all the above field observations were in agreement with former greenhouse-scale results. These indicate the establishment of preferential water fluxes near plants (probably related to the vegetation-induced macropore formation, including at the upper media layer). A question arises then whether non-intentionally introduced but “vigorous” plant species should be removed or kept inside biofilters given their potential influences on the system hydrological performances. From an ecological perspective, it could be assumed that such spontaneous species are best suited to overcome biofilter environmental constraints while maintaining (sometimes enhancing) its hydraulic properties on a short to medium time-scale (from a few weeks to months) according to their functional traits (e.g., below ground root architecture, biomass, and distribution). Nevertheless, still from an ecological perspective, as mentioned before, potentially invasive species should be avoided, and therefore,

Eutrochium dubium and *Solidago rugosa* for instances, which are not invasive, should be preferred compared to other species like *Phalaris arundinacea*. Nevertheless, depending on the implementation context of bioretention systems, such biological control may require substantial weeding efforts (and associated technical costs).

Conclusions

This literature review has shown that bioretention media compaction is a common phenomenon occurring during the first stages of biofiltration system evolution, either under greenhouse or field conditions. The primary choices made during bioretention system conception in regard to size, design, media composition, and plant palette (i.e., structural and functional diversity of plant communities) appear determinant for the hydraulic functioning and bioinfiltration performances. Regarding sizing, a [biofilter infiltration area/impervious catchment area] ratio greater than (or equal to) 2% has often been recommended in order to avoid hydraulic “overload” of the systems. In addition to quick stormwater overflows, an “undersized” biofilter would lead to accelerated media compaction, also favoring overflows on the long-term due to reduced infiltration rates. Some field reports have shown that a bioretention ratio of 1/4 to 1/5 (i.e., an infiltration area (allowing for water infiltration into the subsoil) of 20% in relation to the impervious catchment area) may be particularly effective in reducing at least 90% of the conveyed stormwater runoff volume over a 5-year period. Besides, taking into account column experiments (e.g., (Le Coustumer et al. 2007)) and field observations (e.g., Hart (2017)), a topsoil (or a bioretention media layer including the root compartment) of at least 0.5 m of thickness would be recommended in order to avoid media compaction. Such a media thickness has been shown to be related with root development, which characteristics appear to be of main importance to promote water infiltration. Indeed, this literature synthesis confirmed that vegetation could play a determining role in maintaining bioretention performances, especially infiltration (and drainage) flows at levels comparable to those observed at the early beginning of the device implementation, or in ranges ensuring proper hydrological functioning by limiting the effects of naturally occurring compaction or clogging (i.e., sediment deposits). This overall positive effect of vegetation on infiltration is generally observed in fields and can be explained by their root functional traits. Indeed, “intuitive” trends regarding the growing influence of plant functional type on infiltration rates were confirmed and were as follows: low herbaceous < tall herbaceous < shrubs < trees. Furthermore, results confirmed that among the herbaceous species, “prairie grass” (wild type, “rustic” species mixtures) or more “hardy grass”

type covers are to be preferred over more commercial or “domestic” grass type like turfgrass (which have very shallow roots) in order to efficiently prevent surface clogging or media compaction phenomenon, especially in clay-type or fine-textured media. In fact, species with thick/fleshy and tap root/deep root systems are generally to be favored if infiltration is intended to be the dominant runoff management process. In particular, root traits measured in situ (in infiltrating bioretention systems) showed a positive influence of root mass density (RMD) ranging from 0.1 to 2.2 kg root. m³ on hydraulic conductivity throughout the substrate depth. Moreover, depending on the media composition (sandy or clayey) and the plant growth stage, the root influence can be more or less marked. The development of root systems generally tends to favor infiltration in clayey substrates. Species of comparatively higher specific root length as well as root length and root mass density should be preferred for enhancing infiltration in fine-textured soils. A seasonality effect is usually observed in bioretention systems leading to enhanced infiltration rates in spring and summer compared to fall and winter. This could be explained not only by decrease of water viscosity but also by higher soil drying and biological activity resulting in active bioturbation processes, mineralization of organic matter (including plant debris, root turnover, root exudates...), as well as intense period of vegetative growth with marked influences of root development on media hydraulic properties governing water infiltration and percolation.

Perspectives

Further multi-season field investigations regarding the influence of plant functional diversity, especially root traits, are needed to better understand the long-term influences of vegetation onto bioretention system performances, as well as to ensure that selected species would be resilient to local environmental constraints, leading to perennial vegetation cover and limiting alien species invasion. Additional studies on the multiple benefits that high versus low diversity vegetation communities could provide (regarding bioretention/biofiltration performances, biodiversity and biological activity support) is of main importance considering the growing interest about all the ecological functions and ecosystem services that urban green infrastructures could provide to mitigate climate change and other environmental stressors.

Finally, despite stormwater quality treatment was not the topic of this review, the enhancement of water infiltration rate in conjunction with preferential flows usually reported near vegetation rise the question of their long-term influence on water-soluble pollutant transport, including emerging pollutants such as nanoplastics that are below the macropore size range. Acknowledging that

root functional traits can significantly influence water infiltration and that biophysics-based or machine-learning-based models can help to better understand and predict the effects of plant types and preferential flows along roots. Future works could help to design more sustainable and multi-functional bioretention systems with regard to water quality, possibly relying on trade-offs between finer textured soils to increase soil–water contact time and to remove dissolved pollutants on the one hand, and specific root traits to maintain infiltration and drainage while improving also pollutant retention and dissipation on the other hand.

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Data availability Not applicable. However, as a narrative review article, all the regular papers and research documents used for this review writing are listed, and corresponding references are detailed in the table and reference list of the present paper for the reader to search for them if needed.

Declarations

Consent to participate Both authors (DT and EB) consented to participate to this work.

Consent for publication Both authors (DT and EB) consent to publish this work if accepted.

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